

Errata to “Unbounded field operators in categorical extensions of conformal nets”

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1. Sec. 2.1. In the proof of Lem. 2.5, the phrase “for any $\eta \in \mathcal{H}_j(\tilde{J})$ and $\mu \in \mathcal{H}_i(\tilde{J})$ ” should be “for any $\eta \in \mathcal{H}_j(J)$ and $\mu \in \mathcal{H}_i(J)$ ” (i.e., no tilde). Moreover, in the last line of the computation, one should exchange the roles of μ and η .
2. Sec. 2.2. In the proof of Thm. 2.10, one should choose $\tilde{J}$ to be $\tilde{I}'$ rather than merely clockwise to $\tilde{I}$. This is because at the end of the proof one chooses $\chi \in \mathcal{H}_0(I')$ and considers $R(\chi, \tilde{J})$.
3. Sec. 2.4. In the first paragraph of Def. 2.29, one should add the assumption that for each $\mathfrak{a}_1, \dots, \mathfrak{a}_n \in \mathfrak{H}_i(\tilde{I})$ and $c_1, \dots, c_n \in \mathbb{C}$, the operators

$$\begin{aligned} c_1 \mathcal{L}(\mathfrak{a}_1, \tilde{I}) + \dots + c_n \mathcal{L}(\mathfrak{a}_n, \tilde{I}) : \mathcal{H}_k &\rightarrow \mathcal{H}_i \boxtimes \mathcal{H}_k \\ c_1 \mathcal{R}(\mathfrak{a}_1, \tilde{I}) + \dots + c_n \mathcal{R}(\mathfrak{a}_n, \tilde{I}) : \mathcal{H}_k &\rightarrow \mathcal{H}_k \boxtimes \mathcal{H}_i \end{aligned}$$

are localizable; otherwise Rem. 2.30 may not hold.

4. Sec. 2.7. In Def. 2.42, there is a typo: $\tilde{I}_2$ should be clockwise (rather than anticlockwise) to $\tilde{I}_1$. Also, the second paragraph can be more precise. The revised version of Def. 2.42 (with only one paragraph) is as follows.

Let $\mathcal{E}_{\text{par}}^w$ be a categorical partial extension. Given $\mathcal{H}_{i_1}, \dots, \mathcal{H}_{i_n} \in \mathcal{F}$, $\tilde{I} \in \tilde{\mathcal{J}}$, we define $\mathcal{H}_{i_n, \dots, i_1}(\tilde{I})$ to be the set of all

$$(\mathfrak{a}_n, \dots, \mathfrak{a}_1) \in \mathfrak{H}_{i_n}(\tilde{I}_n) \times \dots \times \mathfrak{H}_{i_1}(\tilde{I}_1)$$

such that $\tilde{I}_t \subset \tilde{I}$ and $\mathfrak{a}_t \in \mathfrak{H}_{i_t}(\tilde{I}_t)$ for each $1 \leq t \leq n$, and $\tilde{I}_s$ is clockwise to $\tilde{I}_t$ whenever $1 \leq s < t \leq n$.

5. ec. 3.3. In the paragraph following the proof of Thm. 3.8, when defining $\pi(A)$ using the decomposition $A = UH$, the operator U should be called a partial isometry rather than unitary.
6. Sec. 3.7. In the last paragraph of the proof of Thm. 3.39, the formula $\dim_{\mathcal{C}_V}(U) = \dim_{\text{Rep}^0(A)}(U)$ is not correct, and needs to be deleted. In the subsequent sentence, the formula

$$D(\mathcal{C}_V)/D(\text{Rep}^0(A)) = \dim_{\text{Rep}^0(A)}(U)$$

citing [KO02] Thm. 4.5 is also citing incorrectly. The correct formula in that paper is

$$D(\mathcal{C}_V)/D(\text{Rep}^0(A)) = \dim_{\mathcal{C}_V}(U)$$

These two mistakes indeed cancel each other, which again implies $D(\mathcal{C}_V)/D(\mathcal{C}_U) = \dim_{\mathcal{C}_V}(U)$ and hence finishing the proof of Thm. 3.39.

7. Sec. 3.7. In the proof of Lem. 3.47, the last sentence “Thus they also hold for $V_\Lambda \otimes V_L$ and hence for V_Λ by theorems 3.38 and 3.39” is a little misleading. The more appropriate one should be “Thus they also hold for $V_\Lambda \otimes V_L$ (by theorems 3.38 and 3.39) and hence for V_Λ ”.
8. Ch. A. In the beginning paragraph, the statement “ $\overline{p_s A^* A} = p_s A^* A p_s$ is bounded and positive” should be “ $\overline{q_s A^* A} = q_s A^* A q_s$ is bounded and positive”.
9. Ch. A. In the proof of Prop. A.3, the sequence q_t does not converges in SOT to 1. To fix the issue, one simply replaces q_t by $q_t + E$ where E is the projection onto $\text{Ker} K$.